

Computational Text Analysis Workshop

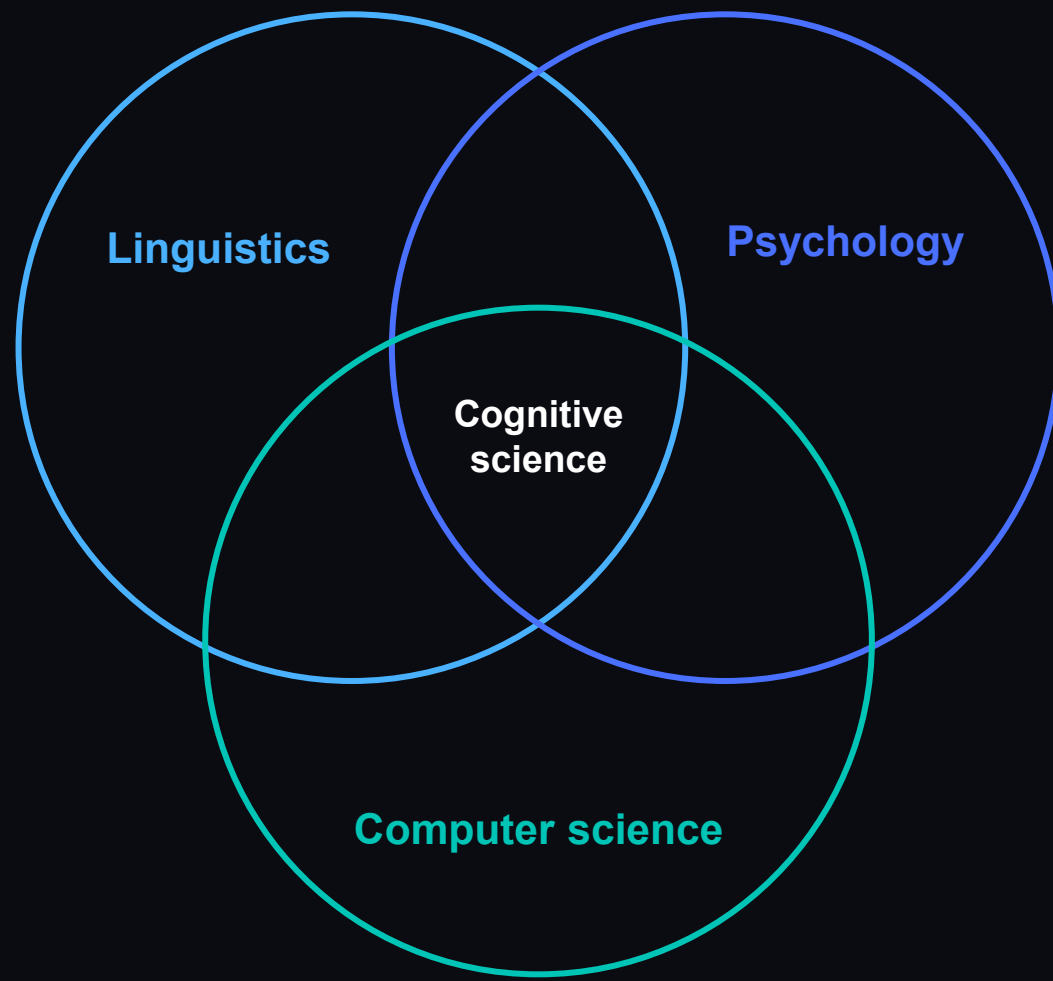
Eli Cochran

University of Kansas

Psychology, Data Science, & Political Science

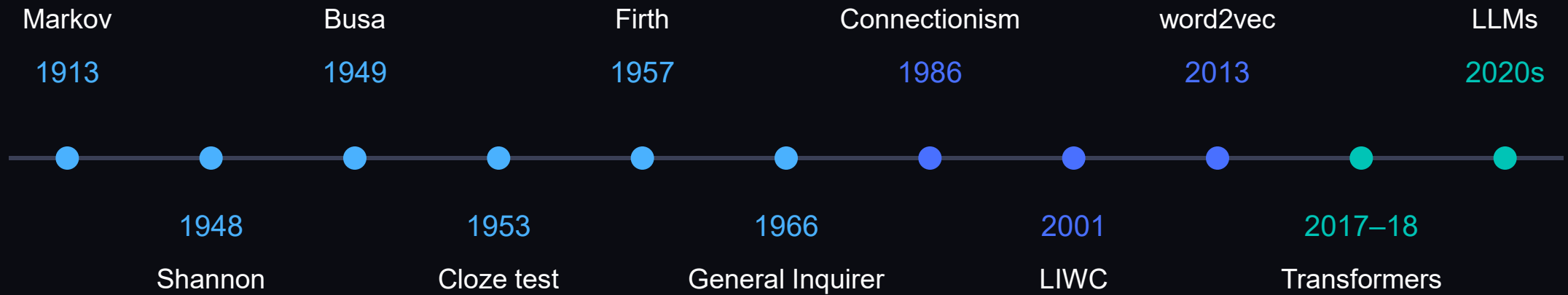
For *reason*...is nothing but *reckoning*
(that is, adding and subtracting) of the
consequences of general names
agreed upon for the marking and
signifying of our thoughts.

- *Hobbes, 1651*



Can meaning be counted?

One hundred years of counting language



1 - Counting

1913-1966 · meaning as frequency

Markov counts letters, 1913

20,000 letters of Eugene Onegin
counted by hand:

A vowel follows a consonant far more
often than chance allows.

So, the next letter depends on the last
one.



● vowel

● consonant

Shannon (1948)

The goalie made a spectacular

S A V ?

Cover the end of a word and guess the next letter.

Bits = a binary measure of entropy

1 vs. 4.7



4.7 bits, no context



1 bit, in context

Father Busa builds the first digital corpus, 1949

Roberto Busa wanted to index Thomas Aquinas's writings.

“How does Aquinas use the word ‘in’?”

A concordance can answer that question.

that God is present

in

all things

the soul is

in

the body

charity is

in

the will

A bag of words keeps the topic and drops the grammar

Lost

dog bites man

man bites dog

Kept

Topic, vocabulary, register

Both are interpreted as {bites, dog, man}.

2 · Filling in the blank

1953 onward · meaning as prediction

Try it: the cloze test, 1953

The goalie made a spectacular _____ in the third period.

I deposited my paycheck at the _____.

The patient's test results came back _____.

The Distributional Hypothesis

**You shall know a word by the
company it keeps.**

J.R. Firth, 1957

3 - Dictionaries

1966 onward · meaning by category

The General Inquirer puts meaning in a table, 1966

Around 180 categories built by hand.

Look every word up simultaneously and count the numbers for each category.



VADER: ratings plus rules

The lexicon

7,500 words rated by hand

love +3.2

okay +0.9

terrible -2.1

The rules

“not good” flips it.

“very good” amplifies it.

“VERY GOOD!!” adds more.

“but” moves the weight to the second clause.

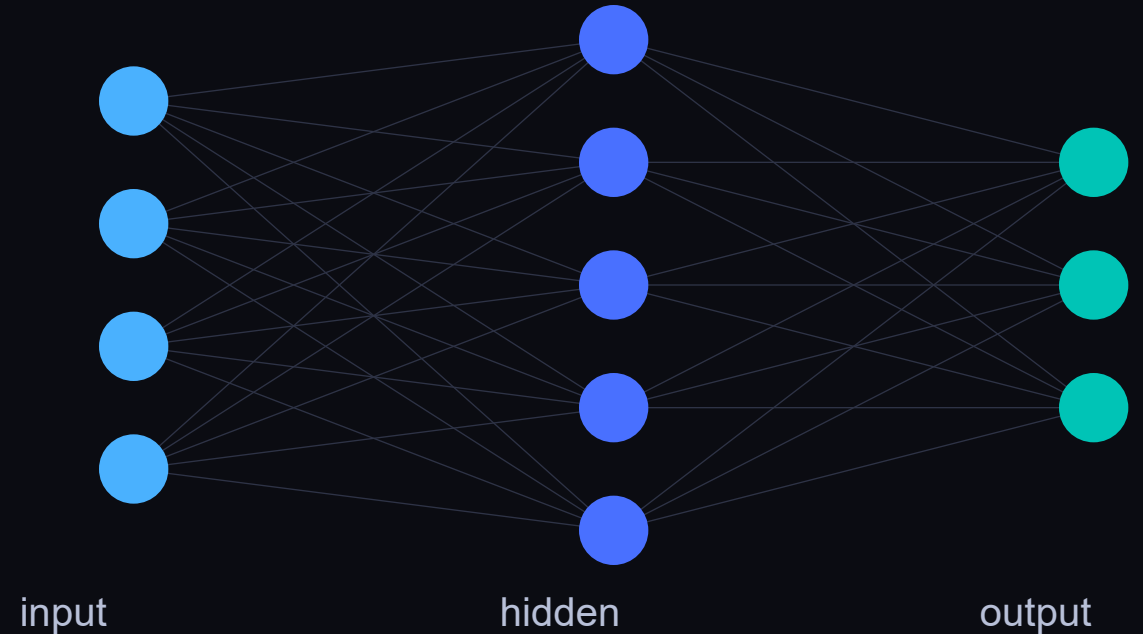
4 · Vectors

1986-2013 · meaning as position

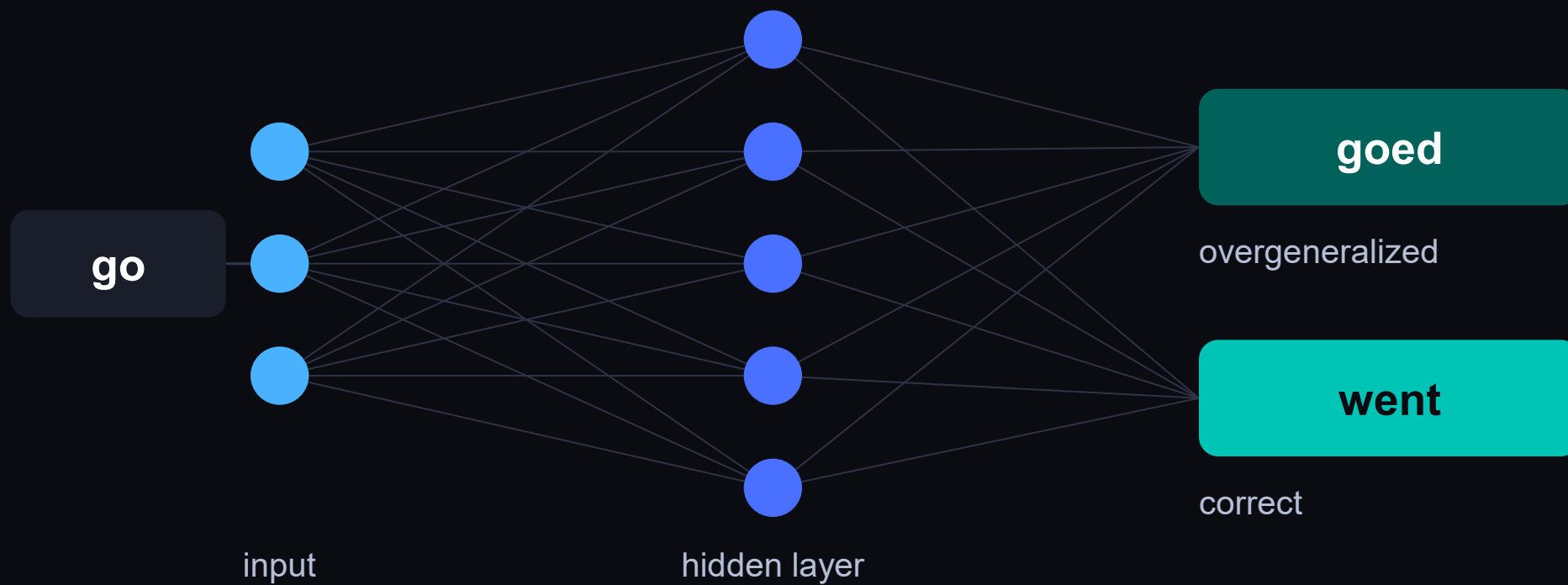
Connectionism: meaning without rules, 1986

Rumelhart and McClelland, 1986. No grammar written down anywhere.

Meaning becomes a pattern of activation, which is a list of numbers.



Learning “went”

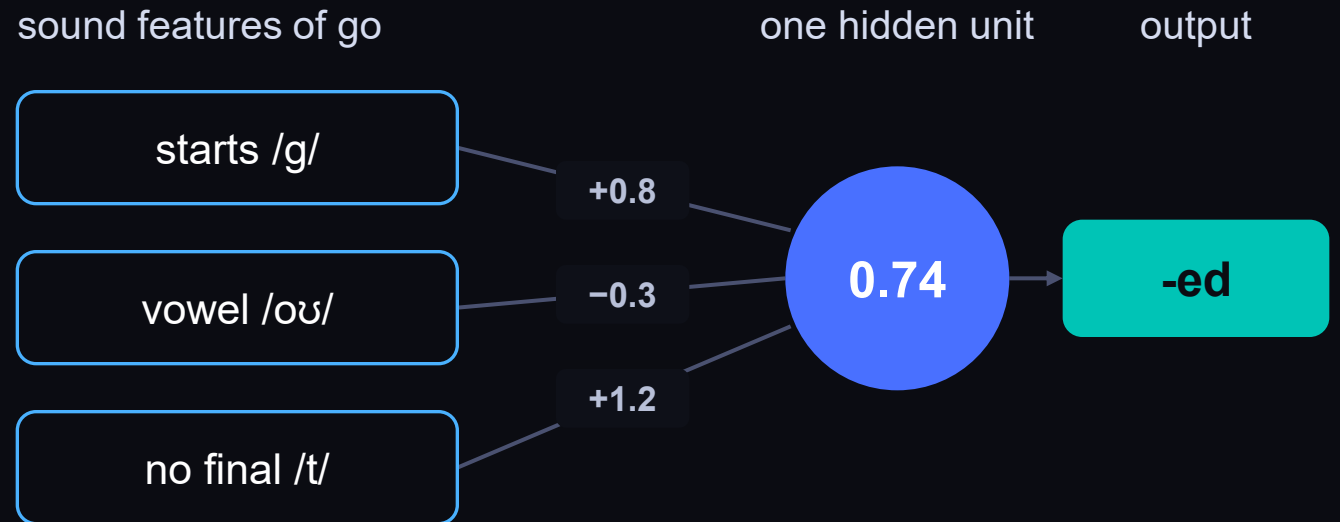


What is the hidden layer doing?

Node: each prediction weight

Edge: how much each feature contributes to the prediction

Learning is just this: adjust your thinking after every mistake.



word2vec: 2013

Predict a word from its neighbours.

Keep only the weights.

Words used in similar cases end up near each other.

king

–

man

+

woman

=

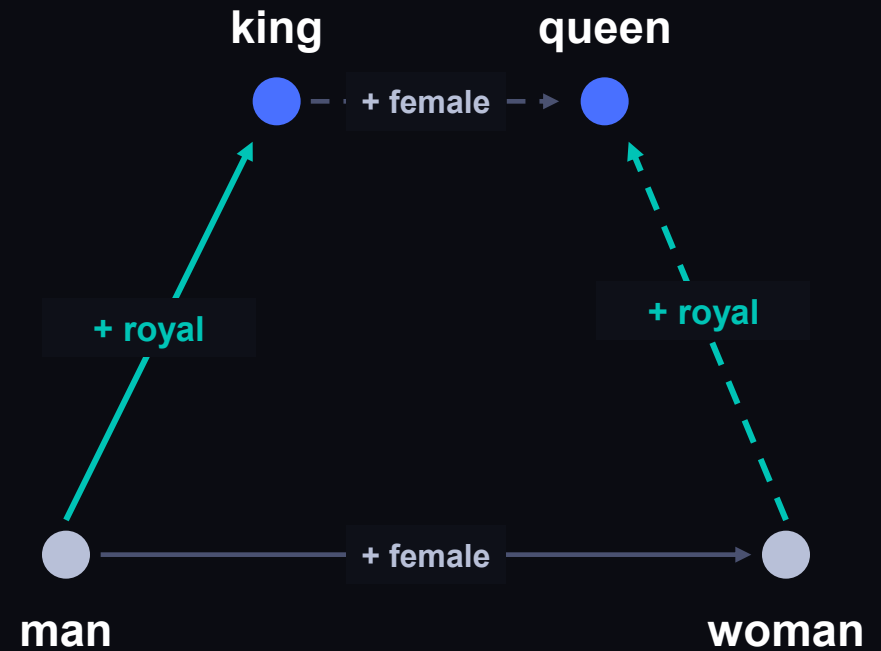
queen

Vectors communicate meaning

Every word is a point.

Words used alike sit close together.

Distance and direction together carry semantic meaning.



5 - Context

2017 onward · meaning in the sentence

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

Lukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin* ‡
illia.polosukhin@gmail.com

Attention: one word, many vectors, 2017

word2vec gives “bank” one vector,
averaging the two meanings

A transformer builds the vector from the
sentence the word is actually in.

I sat on the bank of the *river*.

I *deposited* it at the bank on Main Street.



river sense

money sense



LLMs: more of the same objective

1. Feed it trillions of tokens
2. Transformer-based neural architecture
3. Fluency, world knowledge, instruction following

Workshop time!
